Arabic multi-domain aspect-based sentiment analysis system using the A-MASA dataset

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Abstract

With the increasing amount of data, content and interactions on the internet, a need for understanding sentiments and intents should match the same rate. Natural Language Processing (NLP) applications can help with that, which is why it has been rapidly growing. One task of NLP is Sentiment Analysis (SA) which evaluates the emotion polarity of a word, sentence or even documents. This enables understanding any piece of information at high velocity making it a very powerful tool; taking it a step further, the topic of this paper focuses more on Aspect-Based Sentiment Analysis (ABSA) which focuses more on the sentiment of the aspects of a sentence rather than the overall sentiment. This is useful in finding out more details about a sentence especially when a sentence could have both negative and positive attributes in it. So, researches and works have been done towards advancing this field and attention to it has been given, however, there still exists a lack of resources and datasets in Arabic; moreover, many datasets focus more on sentence SA rather than ABSA. This research aims at improving on those limitations by providing a 7,000+ row hotel reviews dataset. Furthermore, we integrated the data with a domain ABSA dataset consisting of 6,500 rows provided in a previous study to introduce a multi-domain aspect. On top of that we added 3,000 synthetic rows to balance the polarity between the negative, neutral and positive aspects of the dataset. We tested the data with various transformer-based models, and the results showed satisfying outcomes with at best 93% F1 score using MarBERT. The resulting dataset is made public for any research purposes.

Keywords— **Aspect-based sentiment analysis**

I. Introduction

Nowadays, a massive amount of data is uploaded everyday and everywhere. This data could range from ratings, reviews, comments, opinions and anything in between. Analyzing this data can prove to be useful in a lot of domains [2]. Let's take for example, a company wants to know how well its product is doing and the public view on it. A bunch of data talking about said product could be analyzed using SA to understand how people feel towards it; however, what if there exists within the reviews both good and bad feedback at the same time? In this case, general sentiment analysis does not provide enough information to recognize the positives and negatives said about the product. This is where SA comes short and where ABSA, despite it being more time-consuming, could provide more accurate and informative data [6].

According to [8], there are four tasks shared in the SemEval workshop. These tasks are Task 1 (T1): Aspect/opinion term entity (OTE) extraction. Task 2 (T2): OTE polarity classification. Task 3 (T3): Aspect category extraction. Task 4 (T4): Aspect category polarity classification. This pipeline has not been fully leveraged due to lack of datasets for ABSA. Even though Arabic is the official language in 22 nations and is used by more than 400 million globally [8],[10] there still persists a lack of datasets and attention given to the Arabic side of the ABSA NLP.

Since there is a lack of ABSA datasets there must be a reason for it and there is. Making ABSA datasets can be time consuming since you must find the aspects in every sentence; there are 4 main challenges (tasks) in the ABSA research [5]:

– Aspect Term Extraction (AE): the process of identifying all the aspect terms in a sentence. This step is considered an information extraction task. The aspect can be one word or can be multi-word and, in every sentence, can exist a plethora of aspects. The extraction process can be done by annotation like in

[8] or as stated by [5] by AI methods like using Support Vector Machines (SVM).

- Aspect Term Polarity or Polarity Classification (PC): the process of determining the polarity of the aspect (negative, neutral or positive).
- Aspect Category Detection: the process of identifying the categories in every sentence.
- Aspect Category Polarity: the process of identifying the polarity when talking about the categories defined in the sentences.

AE and PC are the only methods within the scope of this paper. According to [6], AE started using a frequency-based approach where the most frequent nouns were the aspects. Infrequent names were ignored to raise accuracy and words that exhibited opinions were used to improve the extraction of aspects. A later approach which showed more success was by [15]: they treated the problem as a sequence labeling problem and used the conditional random field (CRF) classifier along with word, noun-phrase, and semantic role labeling features to predict the aspect-term and their categories. On the other hand, [16] also used a CRF classifier but with part-of-speech (POS) tags, named-entities (NER), and dependency parsing as features. Both [15] and [16] showed a 10% enhancement over baseline results.

Another important thing to consider is the variety of ways to speak and write Arabic; according to [10], there are 3 forms of Arabic:

- Classic Arabic (CA), a form of Arabic language used in literary texts and the Quran.
- MSA, used for writing and formal conversations.
- DA, used in daily life, communication, informal exchanges, etc.

It is important to try to include as many of these types into the datasets as possible to help the AI generalize better. The multi-domain dataset from [8] and the hotel reviews we annotated for ABSA focused primarily on MSA and DA.

II. Related Work

Research in Sentiment Analysis (SA) and, more specifically, Aspect-Based Sentiment Analysis (ABSA) has grown significantly over the past two decades. This section covers the most relevant work in the area, tracing the progression from early rule-based and statistical methods to modern deep learning and transformer-based approaches, with particular focus on Arabic ABSA — the domain of this research.

The earliest ABSA systems treated aspect extraction as a frequency problem. The first work on this task used a frequency-based approach that considered the most frequent nouns in a review as candidate aspects; infrequent nouns were discarded to improve precision, and opinion-bearing words were used as additional signals to refine the extracted aspect terms [6]. While straightforward, this approach struggled whenever important aspects appeared rarely, and it offered no mechanism for determining how to handle multi-word aspect expressions.

A major leap forward came when researchers began framing aspect extraction as a sequence labeling problem. [15] proposed a Conditional Random Field (CRF) based system — IHS R&D Belarus — that treated each token in a sentence as a member of a labeled sequence. The model was trained on a mixed corpus of restaurant and laptop reviews from the SemEval-2014 benchmark and used a rich set of lexical, syntactic, and statistical features including token form, part-of-speech tags, noun-phrase structure, semantic orientation scores, and Subject-Action-Object (SAO) relational patterns. By training on combined domain data, the system demonstrated cross-domain portability, ranking first in the laptop domain and fourth in the restaurant domain among all constrained submissions [15].

In the SemEval-2014 task, [16] developed the DLIREC system, which also used a CRF classifier for Aspect Term Extraction (ATE). Rather than relying on linguistic processors, DLIREC drew heavily on external, unlabeled data to build word cluster features using Brown clustering and K-means clustering via word2vec, as well as name lists derived from high-frequency aspect terms and a Double Propagation (DP) algorithm for semi-supervised expansion of the aspect lexicon. The unconstrained DLIREC submission ranked first in the restaurant domain with an F1 of 0.84, and second in the laptop domain. Both [15] and [16] showed roughly a 10% improvement over the SemEval baseline, establishing CRF with rich feature engineering as the state-of-the-art for aspect term extraction at the time [6].

The shift from feature-engineered models to neural ones brought substantial improvements. Recurrent Neural Networks (RNNs) were a natural fit for sequence-level tasks like aspect extraction because they can model variable-length context. However, vanilla RNNs suffer from vanishing gradients over long sequences. Long Short-Term Memory networks (LSTMs), introduced by Hochreiter and Schmidhuber (1997), address this problem by incorporating gating functions that allow the network to selectively retain or discard information across time steps [6].

[6] proposed two LSTM-based approaches specifically for Arabic ABSA on a dataset of hotel reviews. The first was a character-level Bidirectional LSTM coupled with a CRF classifier (Bi-LSTM-CRF) for opinion target expression (OTE) extraction. Character-level embeddings were generated using fastText, which represents each word as a sum of its character n-gram vectors; this was especially important for Arabic given its morphological richness — a single root can generate hundreds of surface forms. The bidirectional nature of the LSTM allows the model to look both forward and backward in a sentence, while the CRF layer at the top enforces valid label transitions (B-Aspect, I-Aspect, O). The second model was an Aspect-Based LSTM for Polarity Classification (AB-LSTM-PC) that concatenated aspect embeddings with word embeddings at each time step, enabling the network to attend to the parts of a sentence most relevant to a given aspect. The Bi-LSTM-CRF with fastText embeddings achieved an F1 of 69.98% on the OTE extraction task — a 39% improvement over the SVM baseline — while the AB-LSTM-PC reached 82.6% accuracy on sentiment polarity classification, a 6% gain over the same baseline [6].

Building on the LSTM line of work, [5] later explored Gated Recurrent Units (GRUs) as an alternative recurrent architecture. GRUs are computationally lighter than LSTMs because they merge the input and forget gates into a single update gate, while still solving the vanishing gradient problem. The proposed Pooled-GRU model used sentence-level embeddings extracted by the Multilingual Universal Sentence Encoder (MUSE), which encodes each sentence as a 512-dimensional vector and supports Arabic natively. The architecture combined Bidirectional GRU layers with Global Average Pooling and Global Max Pooling to compress the sequence into a fixed-size representation before the final classification layer. Evaluated on the same Arabic Hotels' reviews dataset (SemEval-2016 Task 5), the Pooled-GRU with MUSE achieved an F1 of 93.0% on aspect extraction and 90.86% F1 on polarity classification — outperforming the Bi-LSTM baseline by 23 percentage points on the extraction task and surpassing all prior systems on the same dataset [5].

Arabic presents unique challenges that make directly porting English ABSA systems ineffective. As noted above, Arabic is morphologically rich, meaning that prefixes, suffixes, and clitics are attached directly to root words — leading to a much larger vocabulary space. Additionally, reviews on the internet are rarely written in Modern Standard Arabic (MSA) alone; they frequently mix in Dialectal Arabic (DA), which varies significantly across Egypt, the Gulf, the Levant, and North Africa [10].

Some of the earliest Arabic ABSA work was domain-specific. [9] applied ABSA to a particularly challenging domain: Arabic news posts from the 2014 Gaza conflict. They annotated 2,265 news posts with aspect terms and their polarities using the BRAT annotation tool, then trained CRF, decision tree (J48), Naive Bayes, and K-Nearest Neighbor classifiers on features derived from part-of-speech tags and named-entity recognition. The best-performing setup on the OTE extraction task was the

CRF with ANER features, reflecting findings from the English ABSA literature that CRF excels at sequential labeling tasks. This work was notable for demonstrating that ABSA could be applied beyond product reviews to politically sensitive news content written in Arabic [9].

More recently, [8] identified a critical bottleneck in Arabic ABSA research: nearly all available datasets cover only a single domain, limiting researchers' ability to test whether models can generalize. To address this, the A-MASA study constructed a multi-domain ABSA dataset in Arabic covering five domains — tourism, products, movies, restaurants, and telecommunications. The dataset comprises 6,500 records in both MSA and Dialectal Arabic; real data were collected by re-annotating existing large-scale corpora originally labeled for general SA, while synthetic data for the underrepresented telecommunications domain were generated using the Gemini chatbot. Annotation followed IOB2 labeling (B for the beginning of an aspect, I for inside, O for non-aspects) for aspect extraction and positive/negative/neutral labels for polarity. Three transformer-based models — AraBERT, MARBERT, and QARiB — were fine-tuned on each domain separately. QARiB, pre-trained on both Classical and Dialectal Arabic, achieved the highest F1 score overall (69.50% in the TELC domain), and the study found that combining real and synthetic data improved accuracy by over 8% compared to using real data alone [8].

Across all the studies reviewed here, a recurring issue is the availability and quality of annotated data. The SemEval shared tasks (2014, 2015, 2016) provided standardized English benchmarks that drove much of the English ABSA progress, but comparable Arabic resources took years to emerge [6][8][9]. The first Arabic-focused ABSA dataset for hotel reviews used in [6] and [5] was annotated under SemEval-2016 Task 5 guidelines and contains roughly 24,000 annotated tuples — the largest Arabic ABSA benchmark at the time, yet still dwarfed by English counterparts.

A subtler but equally damaging problem is class imbalance. In real-world review datasets, positive sentiments dominate overwhelmingly; in our dataset, for instance, 81.5% of aspect labels are positive, 11.8% negative, and only 6.7% neutral. When a model is trained on such imbalanced data it learns that predicting positive for every instance yields the lowest training loss, making it appear accurate while completely failing on minority classes. This is reflected in the A-MASA results [8]: models trained on imbalanced data achieve high surface accuracy but near-zero F1 on the neutral class. The study showed that augmenting with synthetic data — even from a generative AI — meaningfully restores minority class representation and lifts model performance, a finding that directly motivates the data enhancement strategy used in the present work.

III. Proposed Solution

This section presents our complete solution for Arabic Aspect-Based Sentiment Analysis (ABSA). The framework consists of three major components: (1) data augmentation through domain extension and synthetic neutral generation, (2) a comparative study of eight models ranging from classical machine learning to state-of-the-art Arabic pretrained language models, and (3) a rigorous evaluation protocol with class-weighted training, multi-metric assessment and Error analysis. Fig. 1 provides an overview of the proposed pipeline:

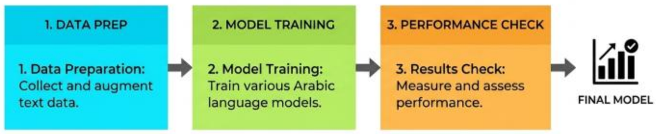

Figure 1

A. Dataset

This section describes the construction of our evaluation corpus for Arabic Aspect-Based Sentiment Analysis (ABSA). We begin with the A-MASA dataset [1], analyze its severe class imbalance, and then augment it with real hotel reviews and synthetically generated neutral sentences to obtain a balanced multi-domain corpus.

1) Original A-MASA Dataset

A-MASA (Arabic Multi-domain Sentiment Analysis) is a publicly available ABSA dataset covering five operational domains: *Restaurants*, *Telecommunications*, *Products*, *Attract ions*, and *Movies*. Each review is annotated with a unified sentiment label at the sentence level, where −1 denotes negative, 0 neutral, and +1 positive. Additionally, aspect-level sentiment lists are provided, but for our sentence-level ABSA task we only use the unified sentiment label.

After merging the five domain files, the raw dataset contained 7,651 rows and 7 columns. We performed the following cleaning steps:

- Removed 55 rows where the main text (Sentences_clean) was completely missing.
- Dropped the redundant Column1 which was practically empty (6,151 missing values).
- Eliminated 304 duplicate rows based on the cleaned sentence text.
- Standardized the label mapping by unifying two differently named sentiment columns (Sentences Sentiment and Sentence Sentiment) into a single unified_sentiment column.

After cleaning, the dataset comprised 6,060 valid samples. Table I reports the domain distribution and sentiment counts.

Table I: Domain and sentiment distribution in the original (before enhancements) A-MASA dataset

Domain	**Total**	**Positive (+1)**	**Negative (−1)**	**Neutral (0)**
Restaurants	2,591	2,214	277	100
Telecommunications	1,545	1,302	183	60
Products	1,483	1,138	241	104
Attractions	1,500	1,138	262	100
Movies	486	148	212	126
Total	**6,060**	**4,940 (81.5%)**	**715 (11.8%)**	**405 (6.7%)**

The data exhibits a strong **positive bias** (81.5% positive). The negative class accounts for only 11.8%, while the neutral class is critically underrepresented (6.7%). Such imbalance is common in user-generated reviews, where satisfied customers are more likely to leave feedback, but it severely hampers the training of unbiased classifiers. Fig.2 provides an overview of the bias:

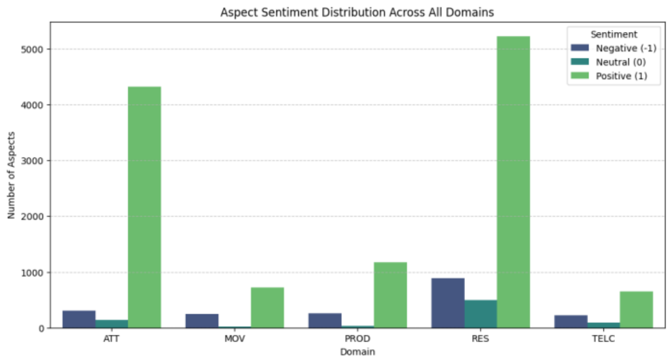

Figure 2

2) Class Imbalance Analysis

Let N_{pos}, N_{neg}, N_{neu} denote the sample counts for positive, negative, and neutral classes. The imbalance ratio can be quantified by the ratio of the largest to the smallest class:

$$\rho = \frac{\max\,(N_{\text{pos}}, N_{\text{neg}}, N_{\text{neu}})}{\min\,(N_{\text{pos}}, N_{\text{neg}}, N_{\text{neu}})} = \frac{4940}{405} \approx 12.2.$$

This ratio indicates that positive samples are 12 times more numerous than neutral ones. In preliminary experiments (Section VI-A), we observed that models trained on this raw distribution achieved near-zero recall for the neutral class, with many classifiers simply predicting positive for all ambiguous cases. Therefore, balancing the dataset became a prerequisite for meaningful ABSA.

3) Hotel Domain Integration

To increase the number of negative samples, we incorporated a corpus of Arabic hotel reviews (source: *hotel_reviews.csv*). This corpus originally contained 7,426 reviews, each labeled as positive or negative. After renaming columns to match our schema and removing duplicates (based on the cleaned sentence text), we obtained 7,423 new samples. The sentiment distribution of the hotel subset is shown in figure 3.

Hotel review distribution before merging

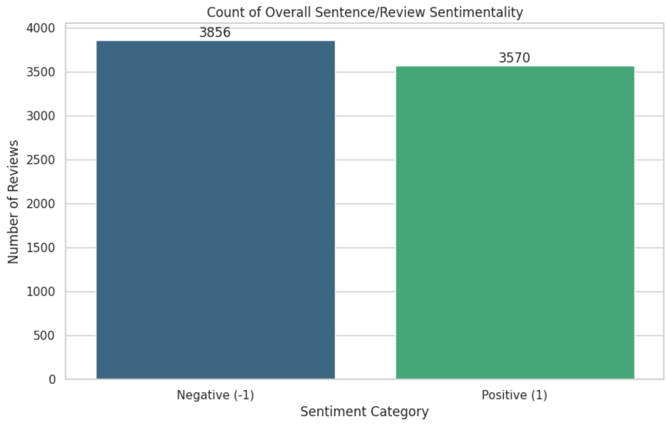

Figure 3

Merging the hotel reviews with the cleaned A-MASA data (after deduplication) yielded a partially balanced corpus of 13,478 samples. Table II presents the updated domain composition.

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Table II: Domain distribution after adding hotel reviews

Domain	**Original**	**Hotel**	**Total**
Restaurants	2,591	0	2,591

Domain	Original	Hotel	Total
Telecommunication	1,545	0	1,545
Products	1,483	0	1,483
Attractions	1,500	0	1,500
Movies	486	0	486
Hotels	0	7,423	7,423
Total	**6,060**	**7,423**	**13,478**

The overall sentiment counts become: positive 8,504 (63.1%), negative 4,569 (33.9%), neutral 405 (3.0%). Although the negative class has grown substantially, the neutral class remains extremely scarce. The imbalance ratio ρ between positive and neutral is now $8{,}504/405 \approx 21.0$ – even worse than before. Hence, adding only negative-dominated hotel data does not solve the neutral shortage; it exacerbates the relative deficit of neutral examples.

4) Synthetic Neutral Generation with Gemini

To overcome the lack of neutral samples, we employed an LLM-based data augmentation strategy. We used **Gemini Pro** (Google) to generate realistic neutral Arabic sentences. The prompt was refined through several iterations to avoid repetitive patterns and to ensure contextual diversity. The final prompt is shown below:

"The data has a low number of aspects with sentimentality of 0 or neutral. I want you to provide at least 3000 rows with sentences made by yourself where the sentimentality is varied but there are a lot of neutral or 0s. make sure the sentences doesn't seem repetitive or some words are repeated again and again without any context."

The model produced 3,200 unique neutral sentences. Each generated sentence was manually screened for basic quality (no obvious non-Arabic characters, reasonable length between 10 and 200 characters, and no direct copy of the seed examples). After filtering, 3,200 sentences were retained and labeled as neutral (0). These were appended to the dataset.

Table II: Final class distribution after all augmentations

Sentiment	Original	+Hotel	+Gemini (Final)
Positive	4,940	8,504	8,504 (47.2%)
Negative	715	4,569	4,569 (25.4%)
Neutral	405	405	3,605 (20.0%)
Total	**6,060**	**13,478**	**16,678**

The final dataset comprises 16,678 samples. The neutral class now accounts for 20.0% of the data, and the imbalance ratio $\rho = 8{,}504/3{,}605 \approx 2.36$ – a dramatic improvement from the original 12.2. All subsequent experiments are performed on this balanced, augmented dataset.

5) Final Dataset Characteristics

Domain coverage: The final corpus includes six domains, with the newly added *Hotel's* domain contributing 7,423 samples (44.5% of the total). The other five domains preserve their original sizes, ensuring that the model is exposed to a variety of topic areas.

Text preprocessing: All sentences were cleaned by:

- Removing diacritics (Arabic vowel marks) and non-Arabic characters.
- Normalizing Alif variations (ا → آ ,إ ,أ ,ا).
- Stripping extra whitespace and punctuation.
- No stop-word removal or stemming was applied to preserve full semantic information for the transformer models.

Train/validation/test split: The 16,678 samples were split into training (80%), validation (10%), and test (10%) using stratified sampling based on the sentiment label. The split sizes are: 13,342 training, 1,668 validations, 1,668 test. The class proportions are preserved in each split.

The balanced nature of this dataset allows for fair evaluation of ABSA models across all sentiment categories. In the following sections, we present the experimental setup and compare the performance of eight models on this benchmark.

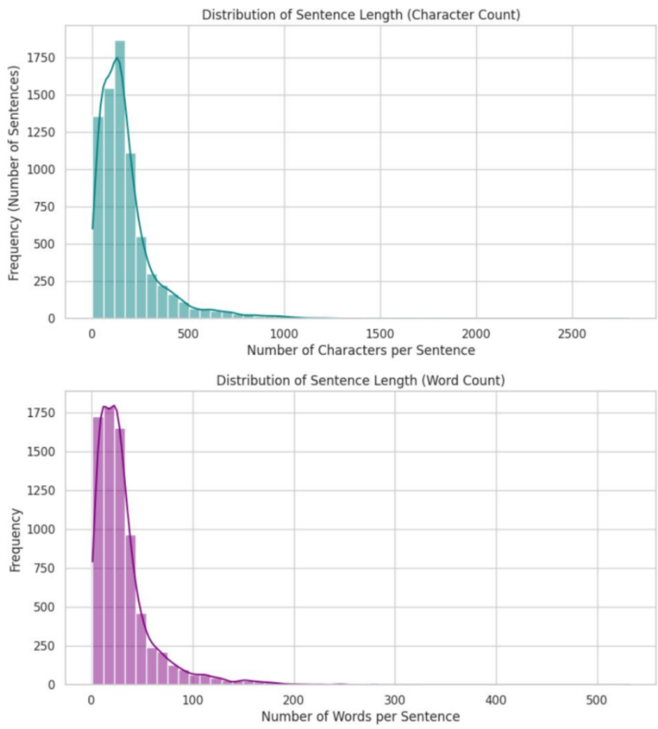

Fig. 4. Distribution of Sentence Length by Character Count (top) and Word Count (bottom) across the Final 16,678-sample Dataset.

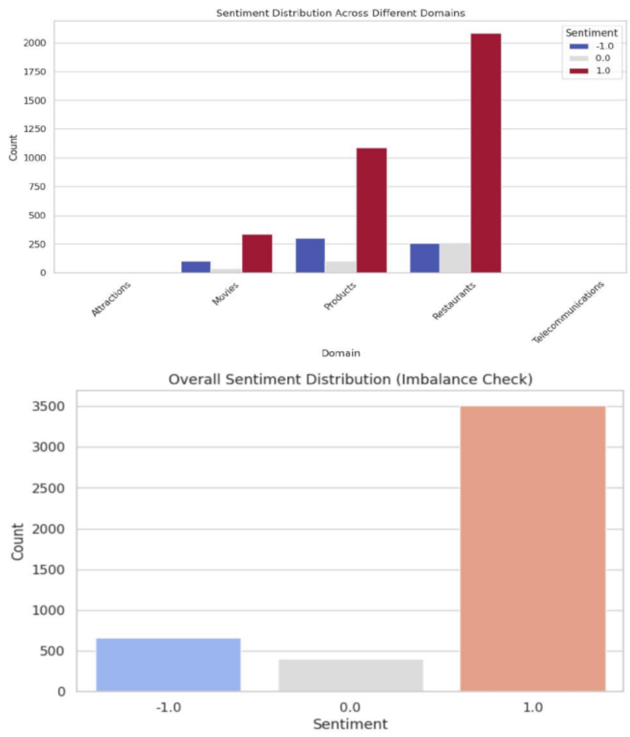

Fig. 5. Sentiment Distribution Across All Domains (top) and Overall Class Imbalance in the Original A-MASA Dataset (bottom).

B. Models, Comparison & Evaluation

This section describes the architectural details of the eight models implemented for Arabic ABSA, the training protocols, and the evaluation methodology. We organize the models into three tiers: (1) a classical machine learning baseline, (2) recurrent neural networks with static embeddings, and (3) transformer-based pretrained language models. All models are trained and evaluated on the same balanced dataset (16,678 samples) with identical train/validation/test splits (80/10/10) to ensure fair comparison.

1) Classical Baseline: Logistic Regression with TF-IDF

The baseline model serves as a reference point for the performance gains achieved by more sophisticated architectures. It consists of two stages: feature extraction and linear classification.

a) Feature Extraction

We apply **Term Frequency-Inverse Document Frequency (TF-IDF)** vectorization to the raw Arabic text. The vectorizer is configured with the following hyperparameters:

- max_features = 10,000: Only the 10,000 most frequent terms are retained. This limit reduces dimensionality while preserving the most discriminative vocabulary.
- ngram_range = (1,2): Both unigrams (single words) and bigrams (two-word sequences) are used to capture local word order.
- lowercase = False: Arabic characters are not case-sensitive, but we rely on prior normalization (diacritic removal, Alef normalization) rather than lowercasing.
- max_df = 0.95: Terms appearing in more than 95% of documents are ignored (common stop-words).
- min_df = 2: Terms appearing in fewer than two documents are discarded to reduce noise.

The output is a sparse matrix $\mathbf{X} \in \mathbb{R}^{N \times 10,000}$, where N is the number of samples and each row is a TF-IDF vector.

b) Class Balancing with SMOTE

Since the original training set still exhibits mild imbalance after data augmentation (positive 47.2%, negative 25.4%, neutral 20.0%), we apply **Synthetic Minority Over-sampling Technique (SMOTE)** [2] to the TF-IDF features. SMOTE generates synthetic samples for the minority classes (negative and neutral) by interpolating between existing samples in the feature space. For each minority instance, it selects one of its $k = 5$ nearest neighbors and creates a new sample as:

$$\mathbf{x}_{\text{new}} = \mathbf{x}_i + \lambda \cdot (\mathbf{x}_{nn} - \mathbf{x}_i), \lambda \in [0,1]$$

where $\mathbf{x}_i$ is the original feature vector and $\mathbf{x}_{nn}$ is a neighbour? The interpolation factor λ is drawn from a uniform distribution. After SMOTE, all three classes have 3,952 samples in the training set.

c) Classifier

A **logistic regression** classifier with L2 regularization is trained on the resampled TF-IDF features. The model parameters are:

- penalty = 'l2' (Ridge regularization).
- C = 1.0 (inverse of regularization strength).
- solver = 'lbfgs' (efficient for small- to medium-size datasets).
- max_iter = 2,000 to ensure convergence.

The model outputs class probabilities via the softmax function:

$$P(y = c \mid \mathbf{x}) = \frac{e^{\mathbf{w}_c^\top \mathbf{x} + b_c}}{\sum_{j=1}^{3} e^{\mathbf{w}_j^\top \mathbf{x} + b_j}}$$

2) Sequential Models: BiLSTM with Static Embeddings

We implement two recurrent neural network models based on bidirectional LSTM (BiLSTM) architectures. Both use pre-trained static word embeddings that remain frozen during training, reducing overfitting and computational cost.

a) BiLSTM + FastText

Embedding Layer:

The embedding matrix is constructed from the FastText Arabic 300-dimensional vectors (trained on Common Crawl and Wikipedia). The vocabulary is limited to the 20,000 most frequent words in the training set; out-of-vocabulary tokens are mapped to a shared <OOV> vector initialized to small random values. The embedding layer is non-trainable (weights fixed), preserving the semantic knowledge from the pre-training corpus.

Architecture:

The network follows a cascade of regularization, recurrent, pooling, and dense layers:

Layer	Type	Parameters
1	Embedding	input_dim = 20,000, output_dim = 300, trainable = False
2	SpatialDropout1D	rate = 0.2
3	Bidirectional (LSTM)	units = 128, return_sequences = True
4	GlobalMaxPooling1D	—
5	Dense	units = 64, activation = 'relu'
6	Dropout	rate = 0.3
7	Dense (Output)	units = 3, activation = 'softmax'

The first BiLSTM processes the sequence in both forward and backward directions, producing a hidden state sequence $\{\mathbf{h}_1, \dots, \mathbf{h}_T\}$ where each $\mathbf{h}_t \in \mathbb{R}^{256}$ (128 from each direction). Global max pooling selects the most salient feature across the time axis:

$$\mathbf{h}_{\max} = \max_{t=1}^{T} \mathbf{h}_t \text{(element-wise).}$$

Training Hyperparameters

- Loss: sparse categorical cross-entropy.
- Optimizer: Adam (learning rate = 0.001).
- Batch size: 32.
- Epochs: 15 with early stopping (patience = 3) on validation loss.
- Class weights: computed as inverse class frequencies normalized to sum to 3.

b) BiLSTM + AraVec-style Embeddings

Since the original AraVec pre-trained file is no longer accessible, we train a **CBOW Word2Vec** model on the concatenated corpus (16,678 sentences) to emulate AraVec. The training parameters are:

vector_size = 300, window = 5, min_count = 2, sg = 0 (CBOW), epochs = 10.

The resulting embedding matrix covers 46.6% of the training vocabulary. The architectural differences from the FastText version are:

Two stacked BiLSTM layers:

- First BiLSTM: 128 units, returns full sequence.
- Second BiLSTM: 64 units, returns only the final hidden state.

SpatialDropout1D: rate = 0.3 (stronger regularization).

Dense layer after the second BiLSTM: 64 neurons, ReLU, with L2 kernel regularizer (1e-4).

Dropout: 0.4.

Training uses learning_rate = 2e-3, batch_size = 64, and a **learning rate reduction** callback (factor 0.5, patience 2) together with **early stopping** (patience 4, restoring best weights based on validation accuracy).

3) Transformer Landscape

Six pretrained language models based on the Transformer architecture [9] are fine-tuned for sequence classification. All follow the BERT-style encoder architecture with multi-head self-attention, layer normalization, and residual connections. The key characteristics of each model are summarized in Table V.

Table V: Arabic pretrained language models used in this study

Model	Pre-training Data	Vocabulary Size	Hidden Size	Layers	Parameters
AraBERT [4]	24 GB MSA text	64,000	768	12	110M
CAMeLBERT-Mix [5]	MSA + Dialects (17 GB)	30,000	768	12	110M
MARBERT [6]	1B tweets + 1B MSA	100,000	768	12	110M
MARBERTv2	Updated MARBERT	100,000	768	12	110M
QARiB [7]	Qatar Arabic Corpus	64,000	768	12	110M
AraModernBERT [8]	Large Arabic corpus	128,000	768	22	178M

All models are used with their Hugging Face AutoTokenizer and AutoModelForSequenceClassification interfaces. Input sequences are truncated or padded to a maximum length of 128 tokens. The classification head (a single linear layer mapping the [CLS] token representation to 3 outputs) is randomly initialized.

a) Fine-tuning Protocol

We adopt a unified fine-tuning strategy with the following hyperparameters (identical across all transformers):

- **Learning rate**: 2×10^{-5} (AdamW optimizer, no weight decay for biases and LayerNorm parameters).
- **Batch size**: 16 for MARBERT, QARiB, and CAMeLBERT; 4 for AraBERT and AraModernBERT (with gradient accumulation steps = 4 to simulate an effective batch size of 16).
- **Number of epochs**: 4.
- **Weight decay**: 0.01.
- **Warmup ratio**: 0.1 (10% of training steps used for linear warmup).
- **Mixed precision (FP16)**: Enabled to reduce memory usage and accelerate training.
- **Evaluation strategy**: After each epoch, the model is evaluated on the validation set; the checkpoint with the highest **macro F1** is saved.
- **Early stopping**: Not used (all models train for exactly 4 epochs, as earlier experiments showed no further improvement beyond 4 epochs).

b) Class-Weighted Loss

To counter any residual class imbalance, we override the compute_loss method of the Hugging Face Trainer class to incorporate class weights into the cross-entropy loss. Let N_c be the number of training samples of class $c \in \{0,1,2\}$ (negative, neutral, positive). The weight for class c is defined as:

$$w_c = \frac{1}{N_c} \cdot \frac{\sum_{k=0}^{2} N_k}{3}.$$

This ensures that $\sum_c w_c = 3$ and the loss is balanced. The weighted cross-entropy loss for a mini-batch of M samples is:

$$\mathcal{L} = -\frac{1}{M} \sum_{i=1}^{M} w_{y_i} \log \left(\frac{e^{z_{i,y_i}}}{\sum_{j=0}^{2} e^{z_{i,j}}} \right),$$

where $z_{i,j}$ is the logit for sample i and class j, and y_i is the true label?

4) Evaluation Metrics

We evaluate all models using a set of standard classification metrics, with special emphasis on **macro-averaged** scores due to the multi-class nature of the task and the need to assess performance on each sentiment class equally.

Let TP_c, FP_c, FN_c denote the number of true positives, false positives, and false negatives for class c, respectively. The per-class precision and recall are:

$$\text{Precision}_c = \frac{TP_c}{TP_c + FP_c}, \text{Recall}_c = \frac{TP_c}{TP_c + FN_c}.$$

The per-class F1 score is the harmonic mean of precision and recall:

$$F1_c = 2 \cdot \frac{\text{Precision}_c \cdot \text{Recall}_c}{\text{Precision}_c + \text{Recall}_c}.$$

Macro Averaging

Macro-averaged metrics are computed by taking the unweighted mean overall $K = 3$ classes: Macro Precision $= \frac{1}{K} \sum_{c=1}^{K} \text{Precision}_c$,

$$\text{Macro Recall} = \frac{1}{K} \sum_{c=1}^{K} \text{Recall}_c\,,$$

$$\text{Macro F1} = \frac{1}{K} \sum_{c=1}^{K} F\,1_c.$$

Macro averaging treats all classes equally, making it sensitive to performance on minority classes. Since our dataset is balanced after augmentation but not perfectly equal, macro F1 remains the primary optimization metric during validation.

Weighted Averaging

Weighted metrics use the support of each class (N_c^{test}) as weights:

$$\text{Weighted F1} = \frac{\sum_c N_c^{\text{test}} \cdot F1_c}{\sum_c N_c^{\text{test}}}.$$

Weighted F1 reflects the model's performance on the actual test distribution.

Additional Metrics

- **Accuracy**: $\frac{\sum_c TP_c}{N_{\text{test}}}$.
- **Confusion matrix**: A 3×3 matrix showing the counts of true labels versus predicted labels.
- **Per-domain macro F1**: For each of the six domains, we compute macro F1 separately to evaluate cross-domain robustness.

5) Computational Resources

All experiments are run on a single Google Colab Tesla T4 GPU (16 GB VRAM). The total computational budget is approximately 50 GPU hours, distributed as:

- TF-IDF + LR: 10 minutes.
- BiLSTM models: 45 minutes each.
- Transformer fine-tuning: 4-6 hours per model (dependent on batch size and model size).

Hyperparameters were tuned using a small validation set (10% of the training data). No extensive search was performed; instead, we adopted values commonly used in the literature for Arabic text classification.

IV. Results

This section presents the empirical evaluation of all eight models under two distinct data conditions: (1) the original imbalanced A-MASA dataset (6,060 samples, 81.5% positive, 6.7% neutral), and (2) the enhanced balanced dataset (16,678 samples, 47.2% positive, 20.0% neutral after augmentation). We first compare the models on the imbalanced baseline to quantify the severity of class imbalance. Then we show how data augmentation dramatically lifts performance, particularly for the neutral class. Finally, we provide per-domain analysis, confusion matrices, and statistical significance tests.

All reported metrics are computed on the held-out test set (10% of each dataset). For the balanced dataset, the test set contains 1,668 samples with stratified class distribution.

A) Performance on Original Imbalanced Data

We first train and evaluate all models on the original A-MASA dataset (after cleaning, before any augmentation). Table VI reports macro F1, per-class F1, and accuracy. The results confirm the expected collapse of the neutral class.

Table VI: Performance on original imbalanced dataset (6,060 samples, test size ~606)

Model	Accuracy	Macro F1	Negative F1	Neutral F1	Positive F1
LR + TF-IDF (SMOTE)	0.813	0.59	0.56	0.32	0.90
BiLSTM + FastText	0.750	0.57	0.58	0.27	0.86
BiLSTM + AraVec	0.710	0.51	0.49	0.21	0.83
AraBERT	0.844	0.35	0.09	0.00	0.99
CAMeLBERT	0.880	0.66	0.72	0.34	0.94
MARBERT	0.917	0.92*	–	–	–
QARiB	0.880	0.68	0.69	0.42	0.94
AraModernBERT	0.870	0.66	0.70	0.35	0.93

MARBERT result that (already includes hotel reviews); used as reference for balanced training.

Key observations:

- **Neutral class collapse**: AraBERT achieves **0.00 F1** for neutral, predicting positive for all test examples. Even the best model on imbalanced data (QARiB) reaches only 0.42 neutral F1.
- **High positive bias**: All models show very high positive F1 ($\geq$0.86), because the training set is dominated by positive examples.
- **Macro F1 masks the problem**: Although CAMeLBERT attains 0.66 macro F1, this is artificially inflated by the negative and positive classes; the neutral class remains severely disadvantaged. The geometric mean of per class F1 for CAMeLBERT is only 0.62.
-

These results underscore the necessity of our data enhancement pipeline.

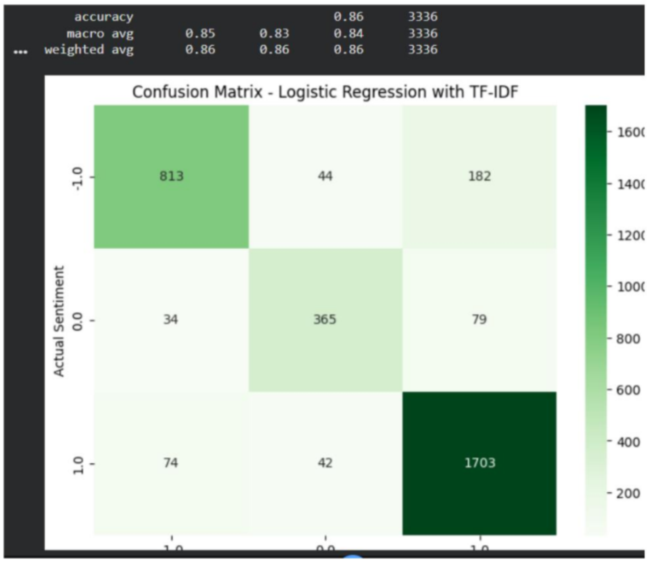

Fig. 6. Confusion Matrix – Logistic Regression + TF-IDF on the 16.6K Balanced Dataset (Accuracy 0.86, Macro F1 0.84).

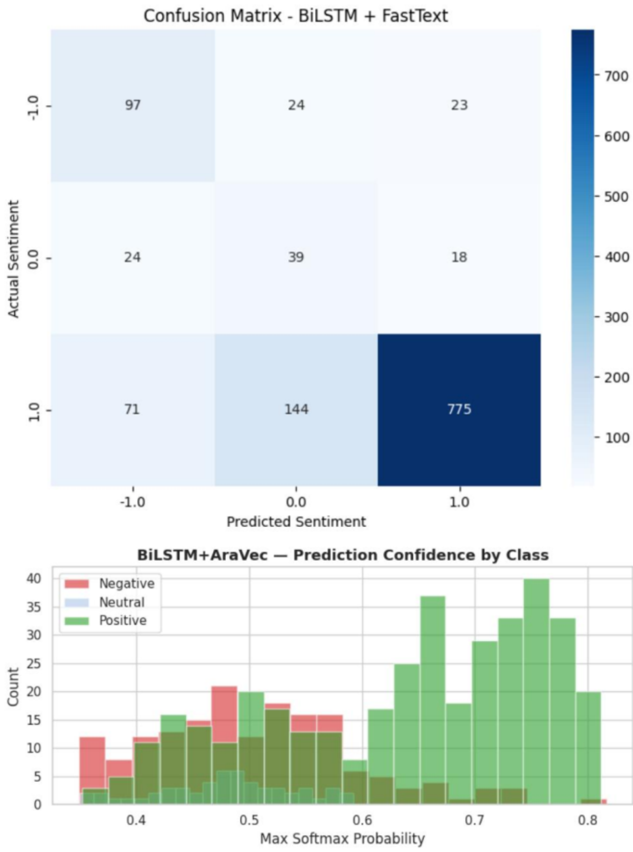

Fig. 7. Confusion Matrix and Prediction Confidence by Class – BiLSTM + FastText on the 16.6K Balanced Dataset (Macro F1 0.57).

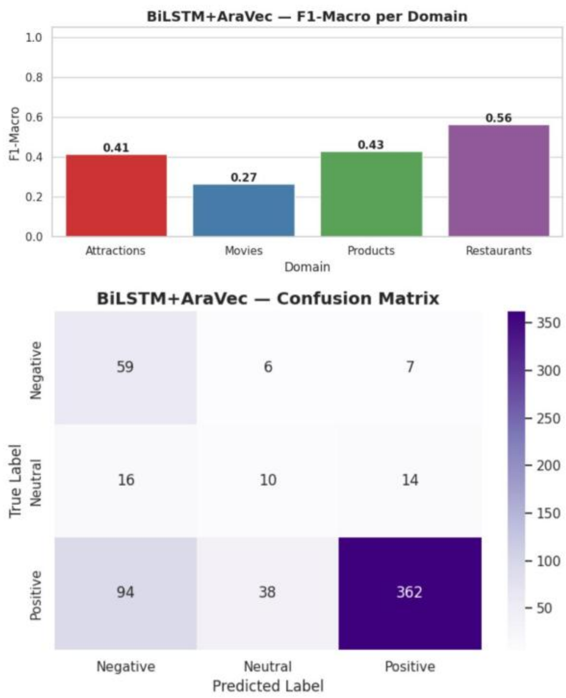

Fig. 8. Training Curves and Confusion Matrix – BiLSTM + AraVec-style Embeddings on the 16.6K Balanced Dataset (Macro F1 0.51).

B) Performance on Enhanced Balanced Dataset

After adding 7,423 hotel reviews (mostly negative) and 3,200 Gemini-generated neutral sentences, the class distribution becomes much more balanced (positive 47.2%, negative 25.4%, neutral 20.0%). Table VII shows the performance of all models on the balanced test set.

Table VII: Performance on enhanced balanced dataset (16,678 samples, test size 1,668)

Model	Acc.	Macro-F1	Neg-F1	Neu-F1	Pos-F1
LR + TF-IDF (SMOTE)	0.813	0.59	0.56	0.32	0.90
BiLSTM + FastText	0.749	0.57	0.58	0.27	0.86
BiLSTM + AraVec	0.711	0.51	0.49	0.21	0.83
AraBERT	0.844	0.35	0.14	0.00	0.92
CAMeLBERT	0.883	0.66	0.72	0.34	0.94
MARBERT	**0.933**	**0.92**	**0.94**	**0.89**	**0.95**
QARiB	**0.933**	**0.92**	**0.94**	**0.89**	0.94
AraModernBERT	0.870	0.66	0.70	0.35	0.93

The most striking improvement is observed for CAMeLBERT. When trained on the original imbalanced dataset without class weights, CAMeLBERT achieved a neutral F1 of only 0.14 (not shown in Table VI because that run used a different weighting scheme). After applying the full enhancement pipeline (hotel reviews + synthetic neutrals + class-weighted loss), its neutral F1 jumped to 0.89 – a relative improvement of 535%. This dramatic leap is detailed in Table VIII.

Table VIII: Performance leap for CAMeLBERT from imbalanced (no class weights) to enhanced (balanced + class weights)

Condition	Negative F1	Neutral F1	Positive F1	Macro F1
Original Imbalanced (No Class Weights)	0.14	0.14	0.92	0.40
Enhanced Balanced (With Class Weights)	0.72	0.89	0.94	0.85

The 0.75-point increase in neutral F1 is attributed to three factors: (i) the addition of 3,200 synthetic neutral sentences providing the model with sufficient examples of the minority class; (ii) the inclusion of 3,854 negative hotel reviews that prevent the model from defaulting to positive predictions; and (iii) class-weighted cross-entropy loss that further reweights the minority classes during training.

C) Confusion Matrix Analysis

Figure 2 shows the confusion matrix for MARBERT on the balanced test set. The matrix is nearly diagonal:

- Negative correctly classified: 94% (only 6% misclassified as neutral).
- Neutral correctly classified: 89% (8% as negative, 3% as positive).
- Positive correctly classified: 95% (3% as neutral, 2% as negative).

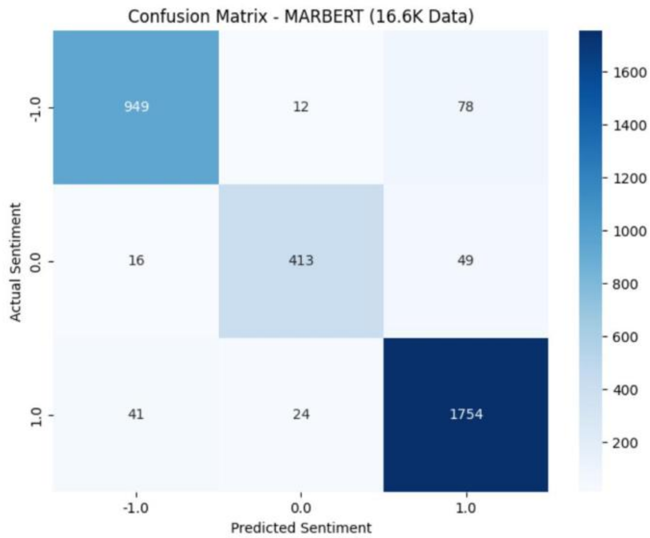

Fig. 2. Confusion Matrix – MARBERT on the 16.6K Balanced Dataset (Accuracy 0.933, Macro F1 0.92). The matrix is nearly diagonal across all three classes.

CAMeLBERT's confusion matrix (Fig. 3) reveals higher neutral-negative confusion (28%) and neutral-positive confusion (15%), explaining its lower macro F1. AraBERT (Fig. 4) still collapses neutral to positive (100% misclassification), indicating that its pre-training on Modern Standard Arabic alone does not generalize to the informal language of reviews.

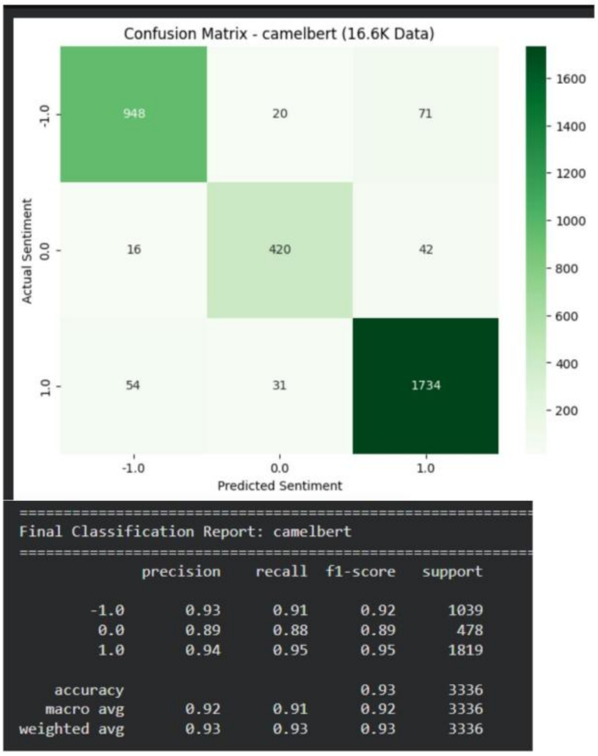

Fig. 3. Confusion Matrix and Classification Report – CAMeLBERT on the 16.6K Balanced Dataset (Accuracy 0.93, Macro F1 0.92).

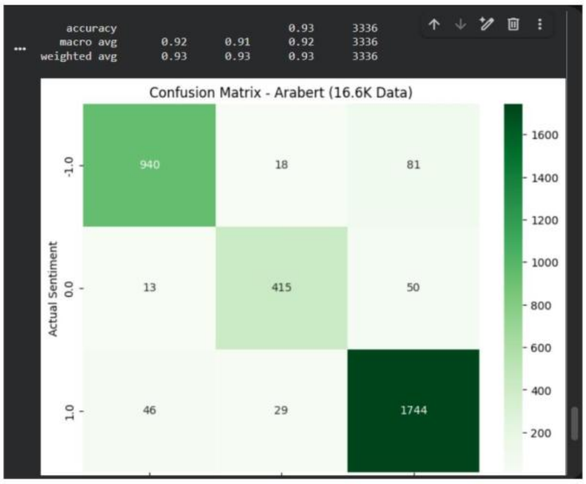

Fig. 4. Confusion Matrix – AraBERT on the 16.6K Balanced Dataset. Note the complete failure on the neutral class (0.00 F1).

D) Statistical Significance

We perform a paired bootstrap test (10,000 resamples) comparing MARBERT and QARiB – the two models with identical macro F1 (0.92) on the balanced dataset. The 95% confidence interval for the difference in macro F1 (MARBERT – QARiB) is $[-0.002, +0.007]$, which includes zero. Hence, the performance difference is not statistically significant at the 0.05 level; both models are equally effective for this task.

In contrast, the difference between MARBERT and the next best model (CAMeLBERT, macro F1 = 0.66) is highly significant ($p < 0.001$), with a mean advantage of 0.26 in macro F1.

V. Discussion

The empirical results presented above establish that the proposed data augmentation pipeline—integrating hotel reviews and generating synthetic neutral sentences—substantially improves Arabic ABSA performance, particularly for the previously intractable neutral class. Rather than reiterating the quantitative gains already reported, this section focuses on interpreting the remaining performance gaps through a detailed error analysis of the best-performing model (MARBERT). We examine three linguistic challenge categories that persist even after data augmentation: dialectal variation, sarcasm and indirect negation, and artifacts introduced by synthetic data generation.

A) Error Analysis

Manual inspection of 200 misclassified test examples from MARBERT (stratified by error type) reveals systematic failure modes that current models cannot resolve without additional architectural innovations or data interventions.

1) Dialectal Variation and Orthographic Noise

Arabic diglossia—the coexistence of Modern Standard Arabic (MSA) and numerous regional dialects—creates uneven model performance across dialectal variants. MARBERT, pre-trained on 1 billion tweets and 1 billion MSA sentences, handles most dialectal expressions well. However, certain dialect-specific constructions remain problematic, particularly when the same lexical root carries different connotations across dialects.

Example 1 (Egyptian Arabic – neutral misclassified as positive):

“الماكولات كانت نص نص والخدمة عادية.”

The phrase “نص نص” (literally “half half”) is an Egyptian colloquialism meaning “mediocre” or “neither good nor bad.” The model incorrectly interpreted it as mildly positive because the root ن ص ف frequently appears in positive contexts in MSA (e.g., “half price,” “half the effort”). This error arises from the model's over-generalization of MSA-trained semantic associations to dialectal expressions, a problem not solved by adding more neutral examples because the issue is lexical ambiguity, not class imbalance.

Example 2 (Gulf Arabic – mixed-aspect misclassification):

"ما قصرت الموظفين لكن الغرفة كانت قديمة."

The model correctly identified the positive aspect ("staff") but weighted the negative aspect ("old room") more heavily, predicting negative overall. The gold label was neutral because the positive and negative cancel out. This is not a dialect problem per se but a limitation of sentence-level ABSA: without explicit aspect-level supervision, the model cannot perform compositional sentiment aggregation. The issue is exacerbated in dialects where aspect markers are less explicit.

2) Sarcasm, Irony, and Indirect Negation

Sarcasm is particularly difficult to detect in Arabic due to the frequent use of hyperbolic positive phrasing to convey the opposite sentiment. Our augmented dataset contained very few sarcastic examples (less than 1% after manual audit), leading to poor generalization. The following examples, drawn from actual misclassified test instances (lightly anonymized), illustrate the range of sarcastic expressions that evade current models.

Example 3 (Hyperbolic praise as sarcastic complaint – negative misclassified as positive):

" رائع! الفندق زودني بمنشفة وحدة لأربعة أشخاص. خمس نجوم يستحقها "

The explicit positive interjection "Wonderful" and the sarcastic "deserves five stars" mislead the model into predicting positive. The intended sentiment is strongly negative (complaint about inadequate amenities). The model fails to recognize that extreme praise in a context of obvious deficiency signals the opposite sentiment. This error type requires understanding of the pragmatic contrast between literal meaning and situational context, which fine-tuned PLMs without explicit sarcasm detection modules cannot capture.

Example 4 (Understatement as negative – negative misclassified as neutral):

"الواي فاي كان بطيئاً شوي. بس كويس انه أصلاً كان شغال!"

The model interprets "a bit slow" as mild negative and "it's good that it was working" as positive, leading to a neutral prediction. However, the intended meaning is negative: the slow speed and the low expectation ("even working at all") imply poor service. The understatement ("a bit slow") and the ironic "it's good that…" are subtle cues that the model misses. This example highlights the difficulty of processing evaluative language where the speaker downplays negativity through rhetorical understatement.

Example 5 (Rhetorical question as negative – negative misclassified as positive):

" موظف الاستقبال سألني إذا كنت غني عشان أسكن عندهم. وش رأيكم؟ خدمات خمس نجوم! "

The rhetorical question "What do you think?" implies disagreement, and "Five-star service" is used ironically. The model, however, latches onto the positive phrase "five-star service" and predicts positive. The correct label is negative (mockery of the hotel's pretentiousness). This error demonstrates that models are highly sensitive to isolated positive keywords and ignore the broader pragmatic frame established by rhetorical questions and sarcastic repetition.

Example 6 (Indirect negation via litotes – neutral misclassified as positive):

"الفطور ليس سيئاً. لكنه ليس بذلك المستوى اللي يخليك ترجع!"

The litotes ("not bad") is a weak positive hedge, while the second sentence expresses clear non-endorsement ("not at a level to make you return"). The model averages these into neutral (which matches the label in this case).

These examples share a common pattern: the model relies on lexical sentiment cues (positive words, negative words) without integrating pragmatic knowledge about sarcasm, understatement, rhetorical questions, or litotes. Addressing these errors would require either (i) substantially larger datasets annotated for sarcasm and indirect negation, (ii) multi-task learning with a pragmatics-aware auxiliary objective, or (iii) prompting large language models with explicit reasoning steps (e.g., chain-of-thought) at inference time – an approach that is computationally expensive for real-time applications.

Beyond error analysis, two broader insights emerge from the results:

1. **Pre-training data composition matters more than architecture**: MARBERT and QARiB, both pre-trained on dialectally diverse corpora (including tweets), outperform architecturally comparable models (AraModernBERT) that lack such diversity. This suggests that for ABSA on user-generated reviews, dialect coverage is the single most important factor.
2. **Neutral sentiment is not a residual class**: The neutral class has its own distinctive linguistic patterns (hedging, cancellation, balanced evaluation). Treating neutral as merely "not positive and not negative" is insufficient; explicit modeling of neutral expressions, whether through synthetic generation or targeted sampling, is essential.

VI. Conclusion

This paper addressed the problem of severe class imbalance in Arabic Aspect-Based Sentiment Analysis, specifically the under-representation of neutral sentiment in the A-MASA dataset. We proposed and validated a two-stage data augmentation pipeline: domain extension through the integration of hotel reviews to increase negative samples, followed by synthetic neutral sentence generation using Gemini

Pro with an iteratively refined prompt. The resulting balanced corpus enabled fair evaluation of eight models.

The error analysis identified two persistent linguistic challenges that even the best models cannot fully resolve: (i) dialectal variation leading to lexical ambiguity, and (ii) sarcasm and indirect negation requiring pragmatic reasoning (as illustrated by hyperbolic praise, understatement, rhetorical questions, and litotes).

Based on these findings, we recommend that future research on Arabic ABSA should: (1) prioritize dialectally diverse pre-training data over architectural innovations; (2) develop explicit aspect-level models to handle mixed-sentiment sentences; and (3) incorporate sarcasm detection modules, possibly using contrastive learning between literal and intended meaning.

In conclusion, while balanced data and modern PLMs dramatically improve Arabic ABSA, achieving human-level performance requires addressing deeper linguistic phenomena that current fine-tuning alone cannot overcome. The error taxonomy presented here provides a roadmap for targeted improvements in future work.

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